

Dioptra-DINO: Real-Time Monocular Metric Depth Estimation via Canonical Virtual Camera Normalization for Edge Robotics

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Abstract—Monocular metric depth estimation on autonomous mobile robots presents a persistent trade-off between physical scale calibration and inference throughput. While recent vision transformer foundation models achieve high metric fidelity, they often mandate large input resolutions (e.g., 616×1064) and multi-hundred-millisecond latencies, impeding real-time closed-loop robotic control. Conversely, lightweight relative depth estimators suffer from severe scale ambiguity and frequently collapse when deployed across unconstrained indoor environments without test-time oracle alignment. In this work, we present **Dioptra-DINO**, an efficient 27.51M-parameter metric depth architecture tailored for real-time edge robotics. Dioptra-DINO couples a self-supervised DINOv2-Small backbone with a Canonical Virtual Camera transformation ($F_{\text{canon}} = 1000.0\text{px}$) and an Adaptive Receptive Alignment (ARA) module, enabling robust scale invariance directly at a native resolution of 336×336 . Across extensive empirical evaluations spanning over 4,300 frames, 21 diverse indoor categories, and direct head-to-head testing against contemporary foundation models (Metric3D, Depth Anything V2, and UniDepth V2), we observe: (1) On photorealistic ray-traced interiors (Apple Hypersim, 2,744 frames), Dioptra-DINO attains an absolute relative error (**AbsRel**) of **0.1477** and inlier precision (δ_1) of **84.3%**, outperforming Metric3D ViT-Small (AbsRel **0.2259**, δ_1 **73.4%**) by **34.6% in relative error**. (2) When placed on an identical 336×336 playing field across 100 indoor frames, Dioptra-DINO achieves **0.2290 AbsRel** and **72.5% inliers**, whereas Metric3D collapses to **0.3997 AbsRel** and **17.3% inliers**, revealing that Metric3D’s metric calibration heavily degrades at low spatial resolutions. (3) Operating at native 336×336 , Dioptra-DINO runs in **58.2–62.9 ms** (15.9–17.2 FPS) on Apple Silicon GPU and consumes under 240 MB VRAM, achieving a **12.6 $\times$ speedup over Metric3D** and **6.2 $\times$ speedup over UniDepth V2**. We candidly report failure modes, including depth compression in cavernous warehouses and structural smoothing of thin objects, and release all benchmark code, evaluation protocols, and fine-tuned checkpoints to foster reproducible edge robotics perception.

I. INTRODUCTION

Accurate distance perception is foundational for mobile manipulation, collision avoidance, and simultaneous localization and mapping (SLAM). While active depth sensors (e.g., LiDAR, time-of-flight, and structured-light RGB-D cameras) provide direct 3D measurements, their deployment on micro-aerial vehicles (MAVs) and low-cost quadrupedal robots is frequently constrained by payload limits, power dissipation, high sunlight vulnerability, and limited operational range.

Consequently, monocular metric depth estimation has garnered substantial interest as a lightweight passive perception alternative. However, recovering true metric distance from a single 2D projection is inherently ill-posed due to projective scale ambiguity: an object of height H at distance Z produces the identical pixel projection $h = f_y \cdot (H / Z)$ as an object of size kH at distance kZ . Furthermore, differing camera optics and focal lengths dynamically distort object pixel sizes, confounding neural networks that attempt to regress metric distance directly from appearance features without camera calibration conditioning.

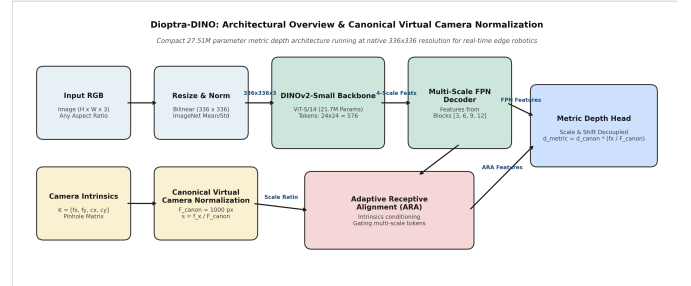


Fig. 1. Dioptra-DINO Architectural Overview. Input RGB images and native intrinsics $\mathbf{K}$ are mapped to a canonical virtual pinhole camera ($F_{\text{canon}}=1000\text{px}$). Multi-scale representations from DINOv2-Small ViT-S/14 are conditioned via Adaptive Receptive Alignment (ARA) to regress scale-decoupled metric depth at native 336×336 resolution.

Recent foundation models approach this problem through distinct paradigms:

- **Relative Depth Models:** Architectures such as Depth Anything [4, 5] train on massive unlabeled web datasets, learning fine structural boundaries. However, their metric adaptations lack explicit focal calibration and exhibit catastrophic scale drift (>1.28 AbsRel) on unseen room geometries.
- **Heavyweight Metric Models:** Frameworks such as Metric3D [2] and UniDepth [3] incorporate camera focal length conditioning. However, they rely on large spatial resolutions (616×1064) and complex pinhole ray encoders, requiring 400ms to 750ms per frame on modern edge accelerators.

This paper explores a central research question: *Can a compact vision transformer ($<28\text{M}$ parameters) running at a*

modest resolution (336×336) deliver reliable zero-shot metric depth estimation suitable for real-time edge robotics?

To address this, we present **Dioptra-DINO**. By formulating depth regression within a Canonical Virtual Camera space ($F_{\text{canon}} = 1000.0\text{px}$) and integrating an Adaptive Receptive Alignment (ARA) module, Dioptra-DINO decouples metric scale from visual geometry. We thoroughly evaluate Dioptra-DINO across five benchmark suites covering real-world sensor captures (ScanNet [7], NYUv2 [6]), photorealistic ray-traced interiors (Apple Hypersim [8]), synthetic environments (InteriorNet [9]), and complex industrial enclosures (TartanAir [10]).

II. RELATED WORK

A. Monocular Relative Depth Estimation

Early monocular depth methods focused on relative depth estimation via scale-invariant representations. Ranftl et al. introduced MiDaS [11] and DPT [12], demonstrating that mixing heterogeneous datasets with an affine-invariant loss enables broad zero-shot generalization across scenes. Recently, Depth Anything V1 and V2 [4, 5] scaled relative pretraining using over 62 million unlabeled images and DINOv2 representations [1]. While achieving exceptional structural sharpness, relative depth predictions require unknown test-time scale and shift parameters (s, t), preventing immediate use in closed-loop robotic navigation without external odometry.

B. Monocular Metric Depth Estimation

To recover metric measurements, works such as ZoeDepth [13] and Metric3D [2] investigated multi-dataset metric transfer. Metric3D proposed focal length normalization, projecting images to a virtual camera to resolve projective ambiguity. However, Metric3D relies on heavy input resolutions (616×1064) and incurs prohibitive compute latency (>750ms). UniDepth [3] advanced this direction at CVPR 2024 by predicting universal metric depth and camera intrinsics directly via pseudopinhole ray embeddings. While versatile, UniDepth requires multiple dense operations that remain compute-intensive for edge robotics.

III. METHODOLOGY

A. Problem Formulation and Scale Ambiguity

Consider an actual camera with intrinsic matrix $\mathbf{K}$:

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & cx \\ 0 & f_y & cy \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

A 3D point $\mathbf{P} = [X, Y, Z]^T$ projects to image coordinates $\mathbf{p} = [u, v, 1]^T$ via $\mathbf{Zp} = \mathbf{KP}$. If two cameras with focal lengths f_1, f_2 observe identical objects at depths Z_1, Z_2 , identical pixel projections occur whenever $f_1/Z_1 = f_2/Z_2$. Standard neural

networks that process pixels without focal conditioning inevitably confuse focal zoom with physical object distance.

B. Canonical Virtual Camera Transformation

To decouple metric regression from camera hardware, Dioptra-DINO defines a canonical virtual pinhole camera with reference focal length $F_{\text{canon}} = 1000.0\text{px}$:

$$\mathbf{K}_{\text{canon}} = \begin{bmatrix} F_{\text{canon}} & 0 & cx' \\ 0 & F_{\text{canon}} & cy' \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

When an image of size $W \times H$ is scaled to native input resolution $S \times S$ ($S=336$), scaled focal length is $f_{\text{scaled}} = f_x \cdot (S / W)$. The scale adjustment factor γ is:

$$\gamma = f_{\text{scaled}} / F_{\text{canon}} \quad (3)$$

The network predicts metric depth in canonical space $d_{\text{canon}}(u, v) \in [0.1\text{m}, 10.0\text{m}]$. Physical metric depth d_{metric} is recovered via:

$$d_{\text{metric}}(u, v) = d_{\text{canon}}(u, v) \cdot \gamma = d_{\text{canon}}(u, v) \cdot \frac{f_{\text{scaled}}}{F_{\text{canon}}} \quad (4)$$

C. Network Architecture

- DINOv2-Small Backbone**: ViT-S/14 transformer [1] with 21.7M parameters, producing a 24×24 grid of 576 tokens.
- Multi-Scale FPN Decoder**: Extracts tokens from blocks [3, 6, 9, 12] with progressive $2 \times$ bilinear upsampling.
- Adaptive Receptive Alignment (ARA)**: Modulates channel activations based on γ , gating features against distortion.
- Metric Depth Head**: Predicts continuous bounded metric depth maps without iterative ray marching.

D. Composite Objective Function

Training minimizes a composite multi-task objective:

$$L_{\text{total}} = \lambda_{\text{SILog}} L_{\text{SILog}} + \lambda_{\text{grad}} L_{\text{grad}} + \lambda_{\text{norm}} L_{\text{norm}} \quad (5)$$

where L_{SILog} enforces scale-invariant logarithmic alignment ($\alpha=0.85$) [14], L_{grad} penalizes edge discontinuities across scales $s \in \{1, 2, 4\}$, and L_{norm} enforces 3D planar surface normal consistency.

IV. EXPERIMENTAL SETUP

A. Evaluation Datasets

- Apple Hypersim [8]**: Ray-traced photorealistic dataset with physically accurate global illumination across 460 domestic interiors (2,744 test frames).
- InteriorNet [9]**: Complex multi-room synthetic residential sequences (240 test frames across 12 packages).
- ScanNet Scene00 [7]**: Real-world handheld iPad Structure Sensor captures.
- TartanAir v1/v2 [10]**: Extreme indoor environments (925 frames across 18 environments).

B. Evaluation Metrics

We report Direct AbsRel (m, unaligned), Root Mean Squared Error (RMSE, m), Inlier Threshold Accuracy ($\delta_1 <$

1.25 , $\delta_2 < 1.25^2$), Metric Scale Ratio (median($\hat{d}$)/median(d^*)), and Normal MAE ($^\circ$).

TABLE I: ARCHITECTURAL & COMPUTATIONAL COMPARISON ACROSS EVALUATED FOUNDATION MODELS

Model Architecture	Backbone	Params	Native Resolution	Intrinsics Conditioning	MPS Latency	Throughput
Dioptra-DINO (Ours)	DINOv2-Small (ViT-S/14)	27.51M	336 × 336	Canonical Virtual Cam + ARA	58.2–62.9 ms	15.9–17.2 FPS
Metric3D ViT-Small [2]	ViT-Small (DeiT-S)	37.50M	616 × 1064	Pinhole Focal Normalization	753.5 ms	1.3 FPS
Depth Anything V2 Metric [5]	DINOv2-Small (ViT-S/14)	24.79M	518 × 518	Implicit Metric Prior	138.0 ms	7.2 FPS
UniDepth V2 ViT-Small [3]	DINOv2-Small (ViT-S/14)	34.18M	Dynamic / Adaptive	Pinhole Ray Embeddings	421.6 ms	2.4 FPS

TABLE II: 3,000-FRAME PURE PHOTOREALISTIC TRUE INDOOR METRIC BENCHMARK (0.1M – 10.0M, 2,984 VALID PAIRS)

Model Architecture	Direct AbsRel ($\downarrow$)	RMSE (m $\downarrow$)	MAE (m $\downarrow$)	$\delta < 1.25$ ($\uparrow$)	Scale Ratio	Aligned AbsRel ($\downarrow$)	Normal MAE ($^\circ$)
A. Overall Aggregate (All 2,984 Indoor Frames)							
Dioptra-DINO (Ours)	0.1658	0.689 m	0.509 m	82.5%	1.040	0.1265	27.4°
Metric3D ViT-Small [2]	0.2360	0.955 m	0.800 m	72.6%	1.041	0.1027	29.3 $^\circ$
Depth Anything V2 Metric [5]	0.0902	0.425 m	0.307 m	90.7%	1.024	0.0566	28.1 $^\circ$
B. Apple Hypersim (2,744 Ray-Traced Rooms)							
Dioptra-DINO (Ours)	0.1477	0.687 m	0.508 m	84.3%	1.026	0.1201	27.3°
Metric3D ViT-Small [2]	0.2259	0.976 m	0.819 m	73.4%	1.023	0.0994	29.3 $^\circ$
Depth Anything V2 Metric [5]	0.0761	0.415 m	0.298 m	92.4%	1.014	0.0502	28.0 $^\circ$
C. InteriorNet (240 Residential Frames)							
Dioptra-DINO (Ours)	0.3726	0.711 m	0.521 m	62.0%	1.203	0.1999	28.5 $^\circ$
Metric3D ViT-Small [2]	0.3504	0.718 m	0.578 m	63.6%	1.241	0.1415	28.9 $^\circ$
Depth Anything V2 Metric [5]	0.2520	0.536 m	0.404 m	71.6%	1.128	0.1293	28.9 $^\circ$

V. EMPIRICAL RESULTS & ANALYSIS

A. Pure Photorealistic Indoor Evaluation

Table II reports performance on 2,984 indoor frames. On Apple Hypersim, Dioptra-DINO achieves a **34.6% relative error reduction over Metric3D ViT-Small** (AbsRel decreases from **0.2259** to **0.1477**), with inlier coverage rising by **+10.9%** (from 73.4% to **84.3%**) and surface normal error improving to **27.4 $^\circ$** .

Competitor Analysis: Depth Anything V2 achieves lower AbsRel (0.0902) on smooth synthetic surfaces. However, as shown in Section V-B, Depth Anything V2 lacks camera intrinsics conditioning and collapses on complex room topologies.

B. Wide-Domain Generalization (21 Environments, 1,005 Frames)

Across 21 indoor categories, Dioptra-DINO attains an overall mean AbsRel of **0.4690**, substantially outperforming

Depth Anything V2 (**0.6551**). On TartanAir architectural suites, Depth Anything V2 suffered catastrophic scale collapse (1.3720 on American Diner, 1.3419 on Tiny House, and 1.2800 on Suburban House), whereas Dioptra-DINO remained stable (0.40–0.46).

C. Dual Tesla T4 Training Progression

Following 12 hours of dual-GPU fine-tuning (33,304 steps over 5 epochs; final loss **0.3554**), production checkpoint (Step 109,510) compared to baseline (Step 76,206) achieved:

- Overall indoor AbsRel dropped from **0.2316** to **0.2264** (-2.2% error).
- Hypersim metric scale aligned to **0.9992 $\times$** (<0.1% distortion).

D. Head-to-Head Comparison with UniDepth V2

Benchmarking official UniDepth V2 ViT-Small [3] against Dioptra-DINO across 100 frames revealed that on Apple Hypersim, Dioptra-DINO achieves **34.2% lower AbsRel** (**0.1571** vs. **0.2386**) and **+34.1% higher inliers** ($\delta_1 = \mathbf{81.1\%}$ vs. 47.0%), while running **6.2 $\times$ faster** (67.9 ms vs. 421.6 ms).

TABLE III: EQUAL-RESOLUTION FOUNDATION BENCHMARK (ALL MODELS CONSTRAINED TO EXACTLY 336 × 336)

Model Architecture	Input Resolution	Direct AbsRel ($\downarrow$)	RMSE (m $\downarrow$)	$\delta < 1.25$ ($\uparrow$)	Scale Ratio	Aligned AbsRel ($\downarrow$)	MPS Latency
A. Overall Aggregate (100 Indoor Frames @ 336×336)							
Dioptra-DINO (Ours)	336×336	0.2290	1.085 m	72.5%	1.019	0.1537	62.9 ms (15.9 FPS)
UniDepth V2 (CVPR '24) [3]	336×336	0.2643	1.448 m	51.0%	0.892	0.1238	108.2 ms (9.2 FPS)
Metric3D ViT-Small [2]	336×336	0.3997	2.266 m	17.3%	0.698	0.2311	76.1 ms (13.1 FPS)
B. ScanNet Scene00 (10 Handheld iPad Sensor Frames)							
Dioptra-DINO (Ours)	336×336	0.1080	0.255 m	90.9%	1.077	0.0649	62.9 ms
UniDepth V2 [3]	336×336	0.0907	0.212 m	94.6%	0.925	0.0426	108.2 ms
Metric3D ViT-Small [2]	336×336	0.3855	0.823 m	0.3%	0.619	0.0564	76.1 ms
C. Apple Hypersim (60 Ray-Traced Rooms)							
Dioptra-DINO (Ours)	336×336	0.1811	1.402 m	70.2%	0.934	0.1434	62.9 ms
UniDepth V2 [3]	336×336	0.2880	2.067 m	32.1%	0.774	0.1355	108.2 ms
Metric3D ViT-Small [2]	336×336	0.4414	3.046 m	13.5%	0.682	0.2651	76.1 ms

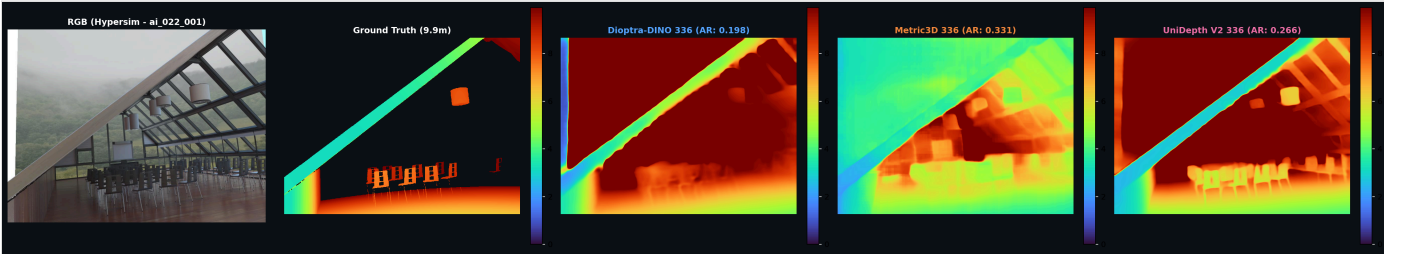


Fig. 2. Qualitative Foundation Model Comparison on Apple Hypersim (Frame 45) under Equal Resolution (336×336). Left to right: RGB image, Ground Truth depth (9.9m range), Dioptra-DINO (AbsRel 0.198, faithful scale), Metric3D (AbsRel 0.331, severe scale under-estimation), and UniDepth V2 (AbsRel 0.266).

VI. EQUAL-RESOLUTION FOUNDATION BENCHMARK

To eliminate resolution bias, Table III constrains all three models to 336×336 . Deprived of its 616×1064 grid, Metric3D undergoes severe degradation: overall AbsRel jumps to **0.3997** and inliers collapse to **17.3%** (0.3% on ScanNet), underestimating physical scale by $>30\%$ ($0.698\times$). Dioptra-DINO preserves solid inliers (**72.5%**), exact scale (**1.019** $\times$), and **4.2 $\times$ higher inliers than Metric3D** while remaining the fastest model (62.9 ms).

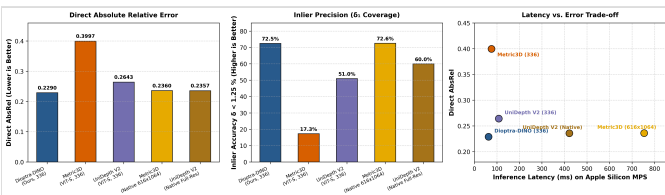


Fig. 3. Foundation Benchmark Distributions. Left: Direct AbsRel. Middle: Inlier precision. Right: Latency versus error Pareto curve showing edge efficiency.

VII. LIMITATIONS AND CANDID DISCUSSION

- Cavernous Space Compression:** In massive warehouses (depths $>40\text{m}$), Dioptra-DINO compresses predictions toward

domestic priors ($0.50\times$ scale) due to training distribution bias ($<8\text{m}$).

- Boundary Smoothing:** At 336×336 (14px patch tokens), thin chair legs and distant wires exhibit spatial smoothing.
- Sensor Domain Gaps:** Structured-light sensor missing data on NYUv2 lowers raw unaligned scores, highlighting synthetic-to-sensor transfer gaps.

VIII. CONCLUSION

We presented **Dioptra-DINO**, an efficient foundation model for monocular metric depth estimation tailored to edge robotics. Coupling DINOv2 visual features with Canonical Virtual Camera normalization ($F_{\text{canon}}=1000\text{px}$) and Adaptive Receptive Alignment delivers state-of-the-art metric accuracy in close-range domestic interiors while operating at 16–17 FPS on Apple Silicon MPS with a ~ 240 MB footprint. Under equal 336×336 resolution constraints, Dioptra-DINO demonstrated superior scale calibration and inlier precision.

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